SOIL MOISTURE CONSERVATION IN COCONUT LANDS

Coconut palm requires continuous supply of water to maintain a consistent nut production throughout the year. The coconut palm has a spreading root system without a tap root. Although the palm has a very large number of roots, their greatest concentration is in the top meter of the soil and within a radius of about 2 m (figure 1).

Each root has a growing point at its end and just behind it there is a whitish region of about 2 mm to 5 mm in length, which absorbs water. The older parts of the root are covered by a thick protective layer which is impervious to water. When the dry condition of the soil persists for a long time, the water absorbing region develops a thick wall resulting in a region which is also impervious to water. Roots so affected remain more or less in a resting condition and cease to absorb water. Therefore, it is essential to give high priority to soil moisture conservation measures during the period without rain.

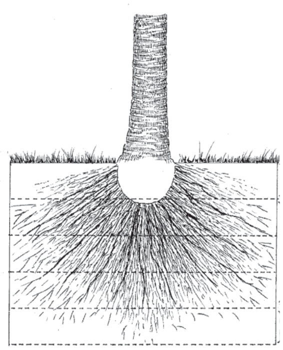

Figure 1: Active root zone of a palm

.The main objectives of soil moisture conservation

- Rain water received to the soil should be stored in the soil.
- Conservation of moisture by minimizing evaporation.

Drought effects

Drought symptoms first appear in seedlings, then in young palms and eventually in adult palms. The activity of the growing point of the crown slows down or may arrests totally due to the drought. Leaf production is reduced resulting in lesser number of inflorescences immerged. Prolonged drought affects the immature nut fall and reduces the number of nuts in a bunch. In addition the size of the nut also reduced. Palms subjected to severe drought take a long period to come into previous productive levels. The first rain received after a long drought is not sufficient to give life to coconut roots. Firstly the inactive growing parts of roots should become active and then the root growth and development of water absorbing regions should take place.

The national coconut production has fluctuated widely during in some years of this decade, primarily as a result of adverse weather conditions. In some years, the loss has been as much as 30%. However, in estates where regular moisture conservation measures are continued obtained a high yields of 10,000 to 15,000 nuts per ha/ year (4,000 to 6,000 nuts per acre/ year).

Soil moisture conservation methods

- Reduction of moisture evaporation from the soil.
- Mulching the manure circle.
- Cultivation of cover crops.
- Adoption of good weed management practices.
- Improvement of water holding capacity of soil.
- Organic matter application to the soil.
- Opening of contour drains
- Burying coconut husks in pits.

1. Mulching the manure circle

Mulch is a common term used to refer to a layer of leaves or similar materials spread over the ground to prevent drying out of soil. Among the materials that could be used as mulch are weed trash, straw and fallen leaves of trees etc. Coconut husks and fronds are the most easily and freely available materials to mulch manure circles in coconut plantations (figure 2).

Benefits of mulching are:

- Reduce moisture evaporation from the soil
- Reduce soil erosion
- Reduce soil temperature during the dry periods
- Suppress weed growth
- Increase organic matter in the soil due to decomposition of mulch materials

1.1 Mulching with coconut fronds

Fallen coconut fronds should be cut into 2-3 pieces and uses them to cover manure circle. Button ends of dried fronds could be used as firewood. Placement of 2-3 layers of coconut fronds is sufficient. With the decaying of old fronds, new mulch could be formed by placing new fallen fronds to cover the area of 2.0 m radius of the palm.

Figure 2:Mulching with coconut fronds

1.2 Mulching with coconut husks

As decaying of coconut husks add different nutrients to the soil it is important as manure. Coconut husks contain high amount of potassium. Approximately, 100,000 husks contain potassium equivalent to 1.0 Mt of Muriate of Potash.

Figure 3: Mulching with coconut husks

Use of coconut husks as mulch

- Single palm requires 200 - 250 husks. Coconut husks should be placed on the manure circle with a 2.0 m radius circle round the palm. Leave 15 cm from the bole of the palm without husks.
- For continuous maintenance of the mulch, add approximately 125 new husks annually.
- Two layers of husks are sufficient. It is not necessary to arrange the husks neatly.
- When fertilizer is applyed remove husk mulch with a mammoty or a rake carefully without disturbing decayed organic layers
- Do not apply husk mulch to palms in water logging areas.
- Palms in water logging areas could be identified by observing root formation on the bottom of trunk above the ground level.

1.3 In addition the above, any other dried plant materials, leaves and weed trash could also be used as mulch.

2. Burial of coconut husks in pits and trenches

For sustainable management of coconut plantations, it is always advisable to recycle husks removed from the nuts harvested to the soil. Generally, husks can absorb water six times of their weight during the wet season and release it slowly to the soil during the dry seasons. The spongy structure of husks provides a good medium for root growth which is an essential character for coconut in hard and shallow soils.

Figure 4: Growth of coconut roots towards the husk pit after one year of burying

Methods of husk burial

Burial of coconut husks in pits

i Cutting pits in between coconut palms.

The size of these husk pits should be 2.4 m long, 1.2m wide and 1.0m deep (8'×4'×3'). These pits should be arranged alternatively between two palms along coconut rows (figure 5 & 6).

Note: In the above method, one hectare of land requires 78 pits (32 pits/ ac) and each pit requires 350-400 coconut husks. The beneficial effects of this method last for about 5 - 6 years. When husks are buried in the second round, open pits at opposite new alternative points (78 pits/ha).

Results of recent experiments conducted in several climatic / soil regimes showed that this method is the best over tested methods detailed bellow.

Figure 5: 8x4x3 feet husk pits

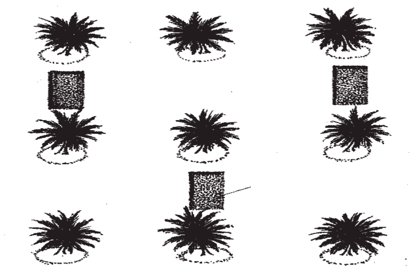

Figure 6: Line diagram of 8x4x3 feet husk pits

ii Individual husk pits for each palm :

i Individual pits can be opened just outside the manure circle, 2 m away from the bole of the each palm of the size 1.2 m long 1.2m wide and 1.0m deep (4' × 4'× 3'). Approximately 200 husks are required for each of these pits (figure 7 & 8).

Figure 7: 4x4x3 feet husk pits

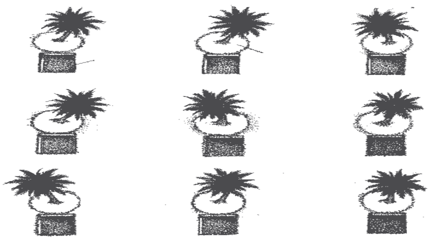

Figure 8: Line diagram of 4x4x3 feet husk pits

iii One circle trenches for each palm

In this method, coconut husks are buried in 0.6m (2 ft) deep and 0.6 - 1.0 m (2 - 3 ft) wide circular trenches covering on area of 1/3 around the palm just outside the manure circle (figure 9 & 10).

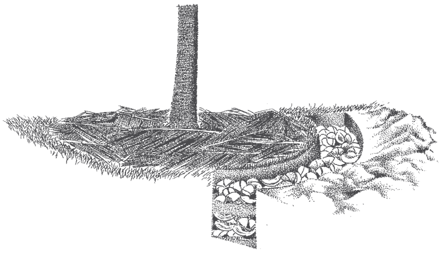

Figure 9: Circular husk pits

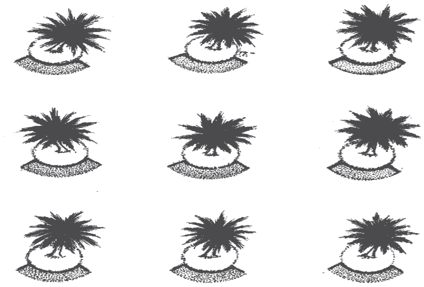

Figure 10: Line diagram of circular husk pits

B) Burying husks in large trenches in between coconut rows

As this method requires a large number of coconut husks, it is suitable only for estates where husks are available excessively. Trenches could be opened along coconut rows beyond the manure circles or across the slop of the land. When a trench is opened in between two rows of coconut, next two rows should be free from a trench. The size of these trenches is 1.0 m wide and 1.0 m deep running along the avenue (figure 11 & 12).

Figure 11: Jumbo husk pits

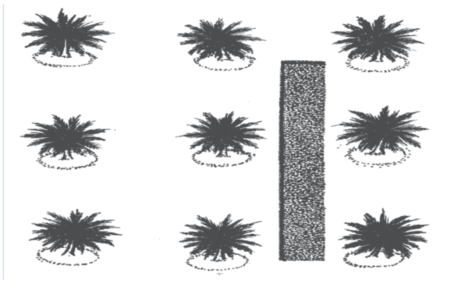

Figure 12: Line diagram of jumbo husk pits

Generally, this method would be ideal for lands where husks are available excessively.

3. Placement of husks in pits and trenches

Husk burying is usually done with the commencement of rains. Husks should be buried in pits or trenches layer by layer, each layer of husks alternating with a layer of soil (10.0 cm), and finely filled up to the ground level. Thereafter, pits / trenches should be closed by piling the remaining soil on top (figure 13 & 14). Systematic arrangement of husks is not necessary as it would not result in any extra benefit.

Figure 13: LS of a husk pit

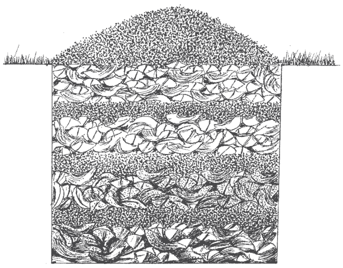

Figure 14: Line diagram of a husk pit

4. Excavation of pits with back-hoe mechines

It is now economical to use 'Back-hoe' machine instead of manual labor for excavating pits / trenches. During this operation, it is important to pile the topsoil close to the pit and use it for filling the pit.

Husk burial is not recommended for waterlogged lands with poor drainage

5. Husk mulching / burial for young palm

Adoption of moisture conservation practices for young palms is also an essential estate practice. At the time of planting seedlings in the field, coconut husks are placed in the bottom of the planting holes.

5.1 Mulching

This can be practiced as detailed for adult palms. However, mulch area should be as follows with the age of seedlings / young palms as shown in table 1.

Age (year)	Distance from the seedlings (meters)	(feet)
1	0.6	2.0
2	0.9	3.0
3	1.2	4.0
4	1.5	5.0
5	1.8	6.0

Note : Maintenance of excess mulch particularly with coconut husks may cause damage by black beetle.

5.2 Husk burial

Individual pits could be established from 2-3 years after planting. Husks are buried in half circular pit prepared beyond the manure circle for each seedling. The depth and width of circular pit are 0.6 m (2 feet) (figure 15). This practice should be continued in every year. After 5 year husk burial should be followed as detailed for mature palms.

6. Other methods which can be practiced to overcome drought effects during dry periods.

- Conserve rain water in coconut lands.
- Plan irrigation properly.
- Mulching should be practiced always to cover the soil.
- Avoid planting of coconut seedlings during dry periods.
- Do not apply inorganic fertilizers.
- Do not practice harrowing or ploughing.
- Manage weeds and minimize the cultivation of intercrops.
- Do not make fires.
- Minimize grazing by cattle.
- Follow monthly coconut picking.

Especially for coconut seedlings.

- Collect and tie leaves of seedlings (figure 15).
- Provide shade.
- Beware of Red-Weevil attack.
- Cut and remove drooping fronds.
- Give a support to protect the crown.

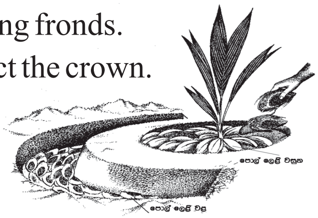

Figure 15: Semicircular husk pits around coconut seedling

Figure 16: Tying of leaves of a coconut seedling

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January 2018